

Vu Anh Duy

AI ENGINEER & COMPUTATIONAL RESEARCHER

1 rue Ernest Renan, 69120, Vaulx-en-Velin

+33 76 67 45 064 (France) | +84 947 686 680 (Vietnam)

Email: avu270201@gmail.com

[LinkedIn](#) - [GitHub](#) - [Portfolio](#)

PROFILE

Computational AI Engineer specializing in deep learning, predictive modeling, and the architecture of complex data pipelines. Combining advanced machine learning expertise with formal certifications in the Science of Gastronomy to bridge the gap between high-dimensional data processing and interdisciplinary food science research.

EDUCATION

Université Claude Bernard Lyon 1

Lyon, France

Master's in computer science, Specialization in Artificial Intelligence (in progress)

Expected August ,2026

- **Core Coursework:** Advanced Machine Learning, Multi-agents' systems, Bio-inspired AI, LLM & LangChain, Image Processing.

INSA Lyon

Lyon, France

Department of Materials Science and Engineering, Engineering Program

2019 - 2022

High School Specializing in Natural Sciences (HSGS)

Hanoi, Vietnam

Very good mention, specialization in Mathematics

2016 - 2019

PROFESSIONAL EXPERIENCE

SNCF Réseau

Saint-Denis, France

AI Research Internship – 3D Data & Deep Learning

Mar 2026 – Sept 2026

- Engineered and trained advanced deep learning architectures (e.g., point-based and voxel-based neural networks) for the 3D semantic and instance segmentation of massive, unstructured LiDAR point clouds, effectively extracting complex contextual relationships from noisy, real-world spatial environments.
- Architected robust, automated preprocessing pipelines for highly dimensional and inherently sparse datasets. Implemented advanced spatial down-sampling, geometric data augmentation, and feature normalization techniques to stabilize model training and manage extreme data variance.
- Constructed large-scale, highly accurate annotated datasets by defining complex structural typologies within dense infrastructure environments. Developed end-to-end data processing workflows incorporating rigorous quality control mechanisms to mitigate class imbalance and ensure high-fidelity ground-truth validation.
- Conducted rigorous quantitative model evaluations using advanced statistical metrics (mIoU, precision, recall, F1-score) to assess predictive reliability. Profiled computational bottlenecks and optimized algorithmic inference times to balance high diagnostic accuracy with efficient, large-scale data processing.

Airmium

Lyon, France

Student Internship

Avril 2024 – Juillet 2024

- Analyzed air quality laboratory data management requirements.
- Proposed and implemented robust architectures for data backup, historization, and secure access, drastically reducing reliance on manual spreadsheet tracking.
- Automated and digitalized processes, reducing manual operations and Excel dependency.

Mạnh Thái Limited Liability Company

Hanoi, Vietnam

Student Internship

Avril 2024 – Juillet 2024

- Managed IT hardware logistics and assembled workstations for local SMEs.

SELECTED PROJECTS

Cyber-Gastronomic Intelligence & Inventory System | *Ongoing*

- **Engineered a homeostatic recommendation system** utilizing custom collaborative filtering and Neo4j knowledge graphs to optimize dynamic food inventory.
- **Digitized qualitative nutritional heuristics** (such as macrobiotic principles, thermal ingredient vectors, and terroir-aware consumption constraints) into quantitative algorithmic rules, fine-tuning Large Language Models to generate balanced, biologically-aligned meal combinations.
- **Deployed lightweight Convolutional Neural Networks (CNNs)** for edge-computing applications to process real-time video feeds during cooking phases.
- **Correlated visual data with physicochemical transformations** by training the computer vision pipeline to monitor the macroscopic effects of thermal processing, structural degradation, and moisture loss in complex food matrices.
- **Designed a predictive machine-learning scoring engine** that aggregates temporal, environmental, and structural variables to forecast the degradation curve of ingredients.
- **Optimized sustainable inventory utility** by prioritizing the consumption of food items at their peak nutritional bioavailability before chemical breakdown occurs.

Factual Reliability Assessment in Large Language Models (KEIGO-SYNC) | *Academic Project*

- Developed an evaluation framework to measure the impact of linguistic registers on the factual reliability of language models.
- Implemented an "LLM-as-a-Judge" architecture (Gemma 3, Llama 3.2, Qwen 2.5) to evaluate semantic accuracy.
- Designed an automated metrics engine utilizing Gemini Embeddings to quantify factual drift and semantic adherence across multiple model architectures (7B to 27B parameters).

Vision Transformer for Image Restoration

- Implemented a 4x4 patch attention pipeline to restore heavily noised images. Conducted comparative performance analysis and authored a comprehensive technical report.

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, LangChain), Java, JavaScript (ES6+), C++, C#, SQL.

AI & Machine Learning: Deep Learning (CNNs, Vision Transformers), Large Language Models (Prompt Engineering, Fine-tuning, LLM-as-a-Judge), Multi-agent Learning (JADE), Bio-inspired AI.

Complex Data Processing: 3D Point Clouds, LiDAR, 3D Semantic Segmentation, PDAL, Data Pipeline Architecture, High-dimensional Data Normalization.

Tools & Environments: Git, GitLab CI/CD, Docker, Jupyter Notebook, Microsoft Azure AI Studio.

CERTIFICATIONS

- IBM Generative AI Engineering (2025)
- Meta Front-End Developer Specialization (2023)
- Google IT Support Certificate (2022)
- Le Cordon Bleu – Art and Science of Multi-Sensory Dining (2025)
- The Science of Gastronomy – University of Hong Kong (2023)

LANGUES

French: professional (C1)

English: professional (C1)

Vietnamese: mother language